

Javascript Assignment

Level – Medium

1. Age Category Checker

```
const age = Number(prompt("Enter your age:"));
if (age < 18) {
  console.log("Minor");
} else if (age <= 60) {

  console.log("Adult");
} else {
  console.log("Senior");
}
```

2. Introduction Formatter

```
const name = prompt("Enter your name:");
const age = prompt("Enter your age:");
const hobby = prompt("Enter your favorite hobby:");
const introduction = `My name is ${name}, I am ${age} years old, and I enjoy
${hobby}.`;
console.log(introduction);
```

3. Sign of Number

```
const n = Number(prompt("Enter an integer:"));
if (n > 0) {
  console.log("Positive");
} else if (n < 0) {
  console.log("Negative");
} else {
  console.log("Zero");
}
```

4. FizzBuzz Enhancement

```

function fizzBuzzArray(arr) {
  for (let i = 0; i < arr.length; i++) {
    const num = arr[i];
    let output = "";

    if (num % 3 === 0 && num % 5 === 0) {
      output = "FizzBuzz";
    } else if (num % 3 === 0) {
      output = "Fizz";
    } else if (num % 5 === 0) {
      output = "Buzz";
    } else {
      output = num;
    }

    console.log(output);
  }
}
const inputArray = [1, 2, 3, 4, 15, 25];
fizzBuzzArray(inputArray);

```

5. Marks Analyzer

```

const N_example = 5;
const marks_example = [80, 90, 75, 60, 95];

const result = marksAnalyzer(N_example, marks_example);

console.log(`Input: ${N_example} students, Marks: [${marks_example.join(', ')}]`);
console.log("Output:");
console.log(`Total: ${result.Total}`);
console.log(`Average: ${result.Average}`);
console.log(`Highest: ${result.Highest}`);
console.log(`Lowest: ${result.Lowest}`);

const N_second = 7;
const marks_second = [76, 64, 67, 94, 87, 88, 91];
const result_second = marksAnalyzer(N_second, marks_second);

console.log("\n");
console.log(`Input: ${N_second} students, Marks: [${marks_second.join(', ')}]`);
console.log("Output:");
console.log(`Total: ${result_second.Total}`);
console.log(`Average: ${result_second.Average}`);
console.log(`Highest: ${result_second.Highest}`);
console.log(`Lowest: ${result_second.Lowest}`);

```

6. Reverse Array

```
const reverseArray = function(arr) {  
  const reversedArr = [];  
  for (let i = arr.length - 1; i >= 0; i--) {  
    reversedArr.push(arr[i]);  
  }  
  return reversedArr.join(' ');  
};
```

```
const input = [1, 3, 2, 5];  
const output = reverseArray(input);
```

```
const input2 = [10, 20, 30, 40];  
const output2 = reverseArray(input2);
```

```
console.log("\nInput:", input2);  
console.log("Output:", output2);
```

7. Sum using Rest Parameters

```
const sumRest = function(...numbers) {  
  const totalSum = numbers.reduce((accumulator, currentValue) => accumulator +  
currentValue, 0)  
  return totalSum;  
};  
const sum1 = sumRest(10, 20, 30);  
console.log("Input: 10 20 30");  
console.log("Output:", sum1);
```

8. Value vs Reference Demo

```
const myArray=[1,2,3];
```

```
function byValue(arr)  
{  
  const newArr=[...arr];  
  newArr.splice(1,1);  
}  
byValue(myArray);  
console.log(myArray);
```

```
function byReference(arr)
```

```
{  
  arr.splice(1,1);  
}  
byReference(myArray);  
console.log(myArray);
```

9. Array Transformation using HOFs

```
const N = 5;  
const inputData = [2, 4, 6, 1, 5];  
const finalOutput = processArray(inputData);  
console.log(`Input Array (${N} elements): [${inputData.join(', ')}]`);  
console.log("Output: The numbers squared and filtered (> 25) are:");  
console.log(finalOutput);
```

10. Custom Calculation with Callback

```
const calculate = function(arr, callback, operation) {  
  const result = callback(arr, operation);  
  return result;  
};  
const inputArr = [2, 4, 6, 8];  
const N_input = 4;  
  
const operation1 = "sum";  
const result1 = calculate(inputArr, performCalculation, operation1);  
console.log("Input:", N_input);  
console.log(inputArr.join(' '));  
console.log(operation1);  
console.log("Output:", result1);
```